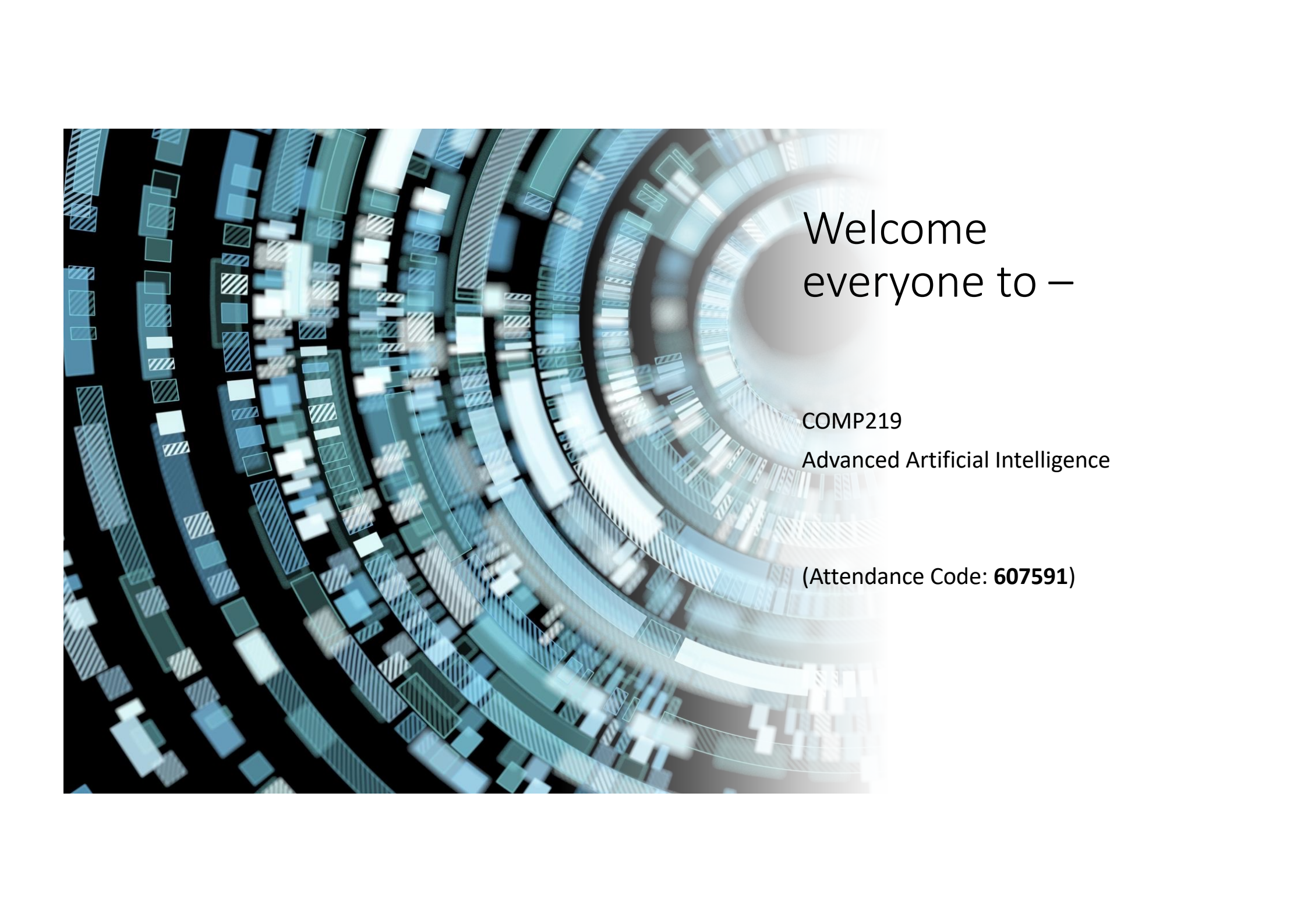


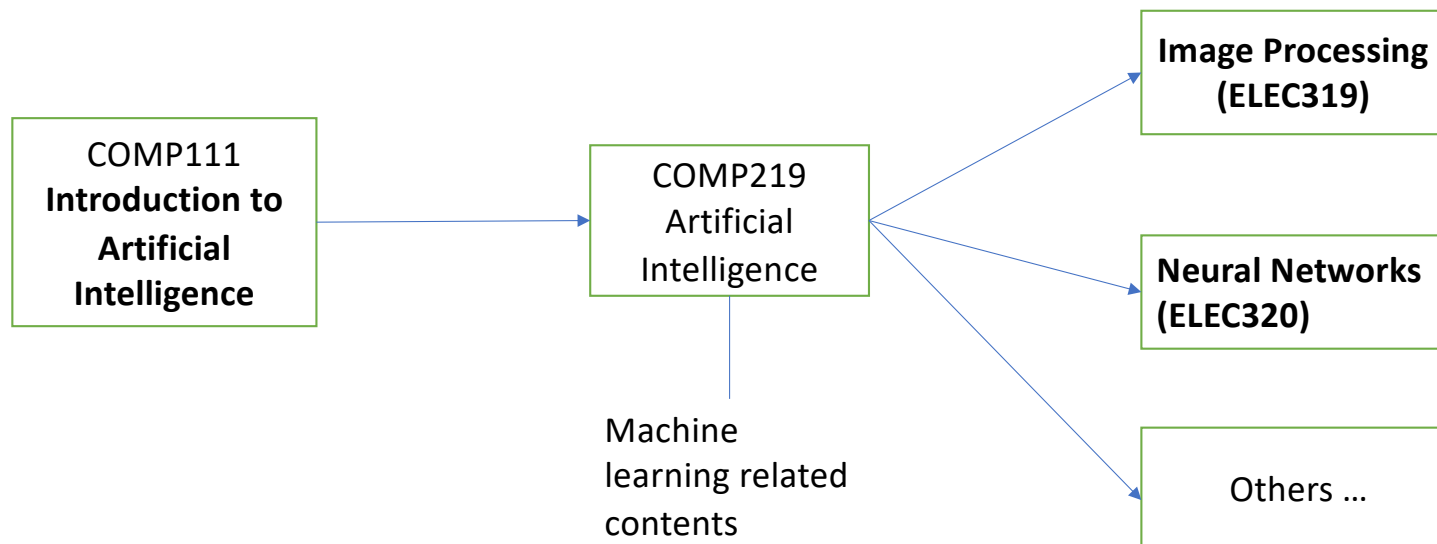


Welcome
everyone to –

COMP219
Advanced Artificial Intelligence

(Attendance Code: **607591**)

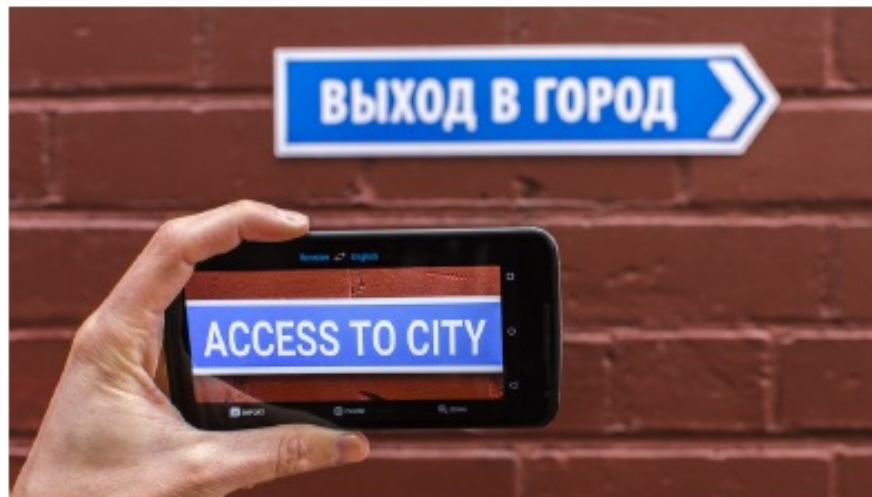
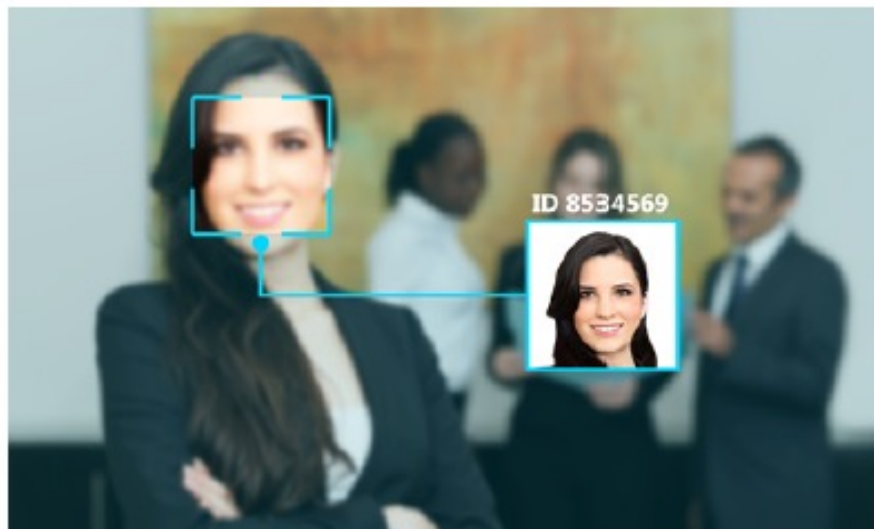
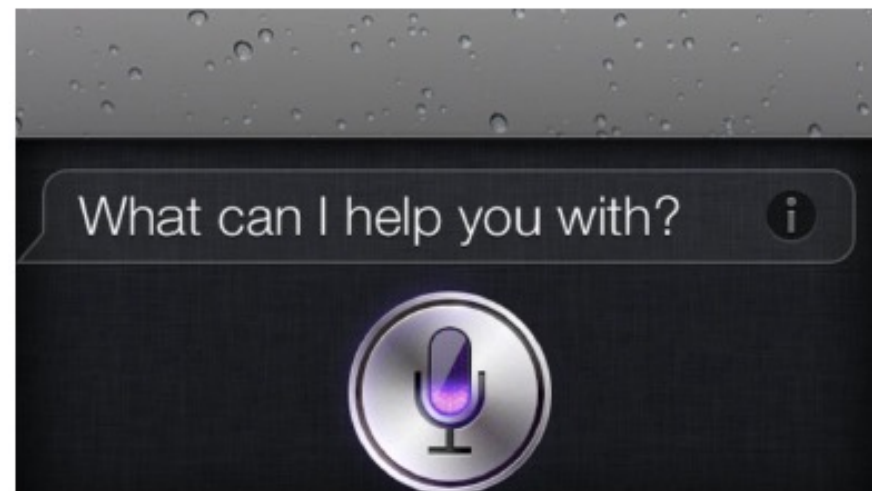
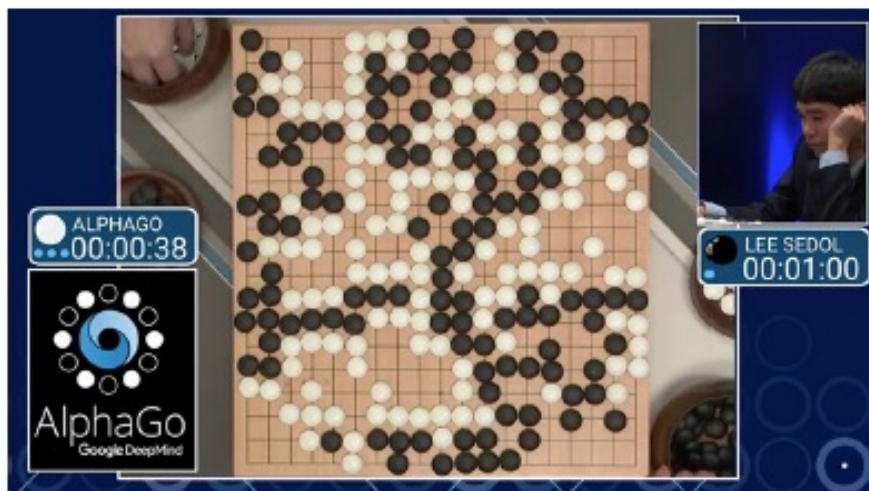
- My website: <https://cgi.csc.liv.ac.uk/~xiaowei/>
- My lab's website: <https://cgi.csc.liv.ac.uk/~acps/home/>



Introduction to COMP219

Prof. Xiaowei Huang

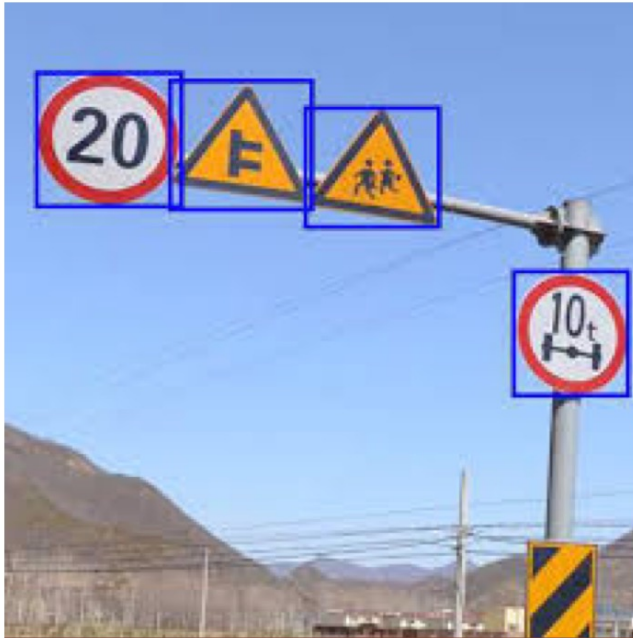
<https://cgi.csc.liv.ac.uk/~xiaowei/>











Risks – security problem?

— AI can create synthetic fingerprints that fool biometric scanners

DeepMasterPrints can dupe systems that only compare partial prints.

<https://www.engadget.com/2018-11-16-ai-fingerprints-biometric-scanners.html>

Risks – privacy problem?

— The Pentagon has a laser that identifies people by their heartbeat

You can't disguise your heartbeat.



Rachel England, @rachel_england
June 27, 2019

33

Comments

40

Shares



<https://www.engadget.com/2019-06-27-the-pentagon-has-a-laser-that-identifies-people-by-their-heartbe.html>

Risk – bias?



Timnit Gebru ✓
@timnitGebru

I'm sick of this framing. Tired of it. Many people have tried to explain, many scholars. Listen to us. You can't just reduce harms caused by ML to dataset bias.



Yann LeCun @ylecun

ML systems are biased when data is biased.
This face upsampling system makes everyone look white because the network was pretrained on FlickrFaceHQ, which mainly contains white people pics.
Train the **exact** same system on a dataset from Senegal, and everyone will look African. [twitter.com/bradpwyble/sta...](https://twitter.com/bradpwyble/status/1271444444444444444)

10:00 PM · Jun 21, 2020



2K

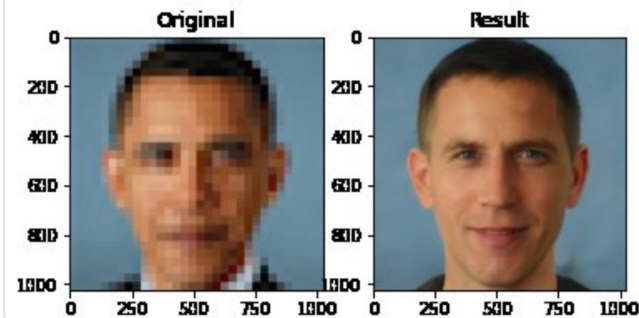


568 people are Tweeting about this



Chicken3gg @Chicken3gg

Replying to @tg_bomze



5:36 PM · Jun 20, 2020



1.9K



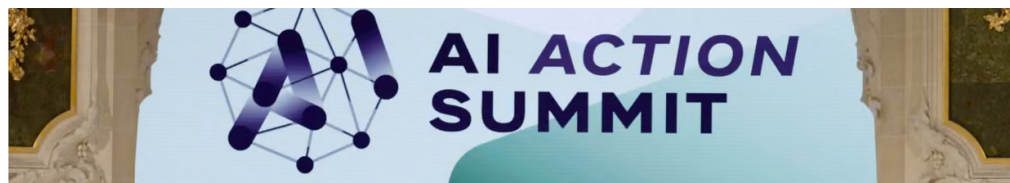
651 people are Tweeting about this

When a blurred image of Barack Obama was fed to the model, it blurred out an image of a Caucasian male. Given the situation right now in the US, it didn't take long for the people to come down heavy on the lack of diversity in ML



US and UK refuse to sign Paris summit declaration on 'inclusive' AI

Confirmation of snub comes after JD Vance criticises Europe's 'excessive regulation' of technology



AI Tug-of-War: Trump Pulls Back Biden's AI Plans

By Adam Aft, Caroline Burnett, Cynthia Cole, Brian Hengesbaugh, Keo McKenzie, Justine Phillips, Elizabeth Roper & Avi Toltzis on January 25, 2025

POSTED IN [ARTIFICIAL INTELLIGENCE](#), [DATA PRIVACY](#), [US](#)

 Listen to this post

On January 20, 2025, the first day of his second term, President Trump revoked [Executive Order 14110](#) on Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence (the "Biden Order"), signed by President Biden in [October 2023](#). In doing so, President Trump fulfilled a campaign pledge to roll back the Biden Order, which the 2024 Republican platform described as a "dangerous" measure. Then on January 23, 2025, President Trump issued his own Executive Order on AI, entitled [Removing Barriers to American Leadership in Artificial Intelligence](#) (the "Trump Order"). Here, we examine some of the practical implications of the repeal and replacement of executive orders by Trump and what it means for businesses.

Nvidia boss says UK will be 'AI superpower' as tech firms invest billions

17 September 2025

Share  Save 

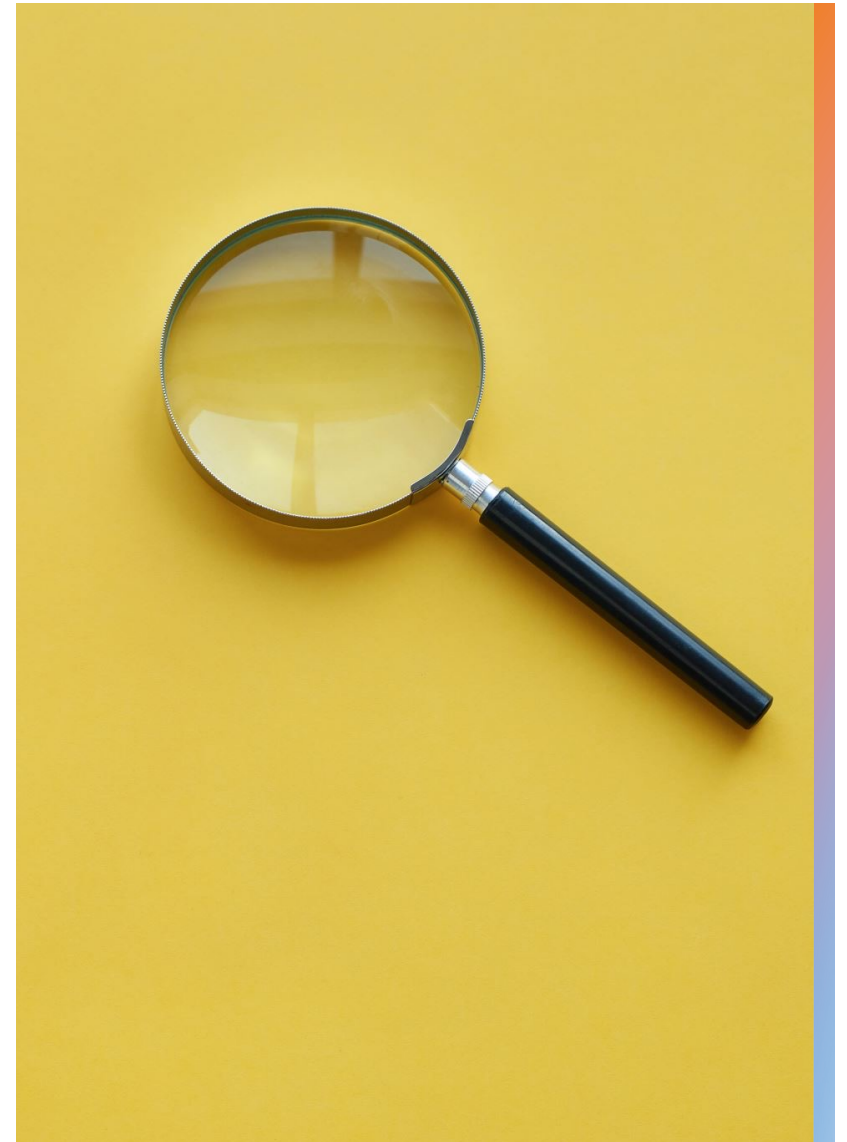
Zoe Kleinman Technology editor and **Faisal Islam** Economics editor



Many apps are AI-driven.

But to know how to create them (**in a responsible way**), it is nothing easy.

This module is to establish a foundation for you to be able to do them in the future.



Pre-requisite Knowledge - Probability

Number of Orders per Week x_i	Probability p_i
41	.03
42	.10
43	.15
44	.17
45	.25
46	.15
47	.10
48	.05

[RAIN/BARK DISTRIBUTION]			
	Rains	Doesn't rain	
Dog barks	9/48	18/48	27/48
Dog doesn't bark	3/48	18/48	21/48
	12/48	36/48	48/48

(1)

Why important?

(2)

Machine learning models are **probabilistic** and perform in an **uncertain (probabilistic)** world.

Pre-requisite knowledge – linear algebra

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix}$$

(1)

Why important?

Machine learning models are
stored and operated in matrix form,

$$x = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} * \begin{pmatrix} 9 \\ 8 \\ 7 \end{pmatrix}$$

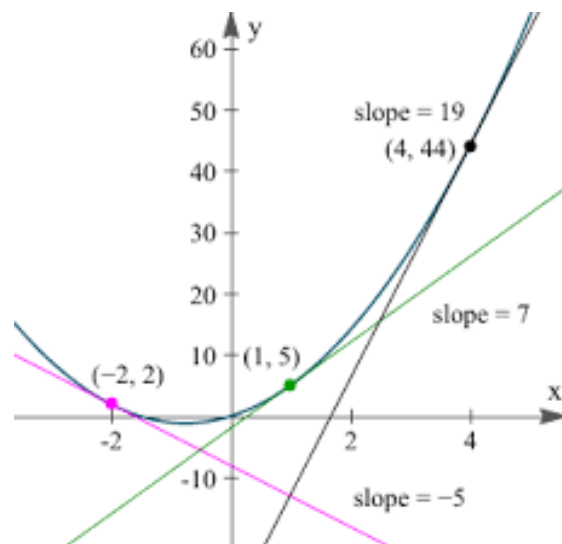
$$1*9 + 2*8 + 3*7 = 46$$

$$4*9 + 5*8 + 6*7 = 118$$

$$x = \begin{pmatrix} 46 \\ 118 \end{pmatrix}$$

(2)

Pre-requisite knowledge – derivative and partial derivative



(1)

Why important?

Training and analysing machine learning models rely on **optimisation** algorithms, which in turn rely on **gradients** (multi-dimensional partial derivative)

$$\begin{aligned} z &= 3x^2 + 2xy - y^2 \\ &= 3(1)^2 + 2(1)(2) - (2)^2 \\ &= 3 \end{aligned}$$

$$\begin{aligned} \frac{\partial z}{\partial x} &= 6x + 2y = 6(1) + 2(2) = 10 \\ \frac{\partial z}{\partial y} &= 2x + 2y = 2(1) + 2(2) = 6 \end{aligned}$$

(2)

Warning



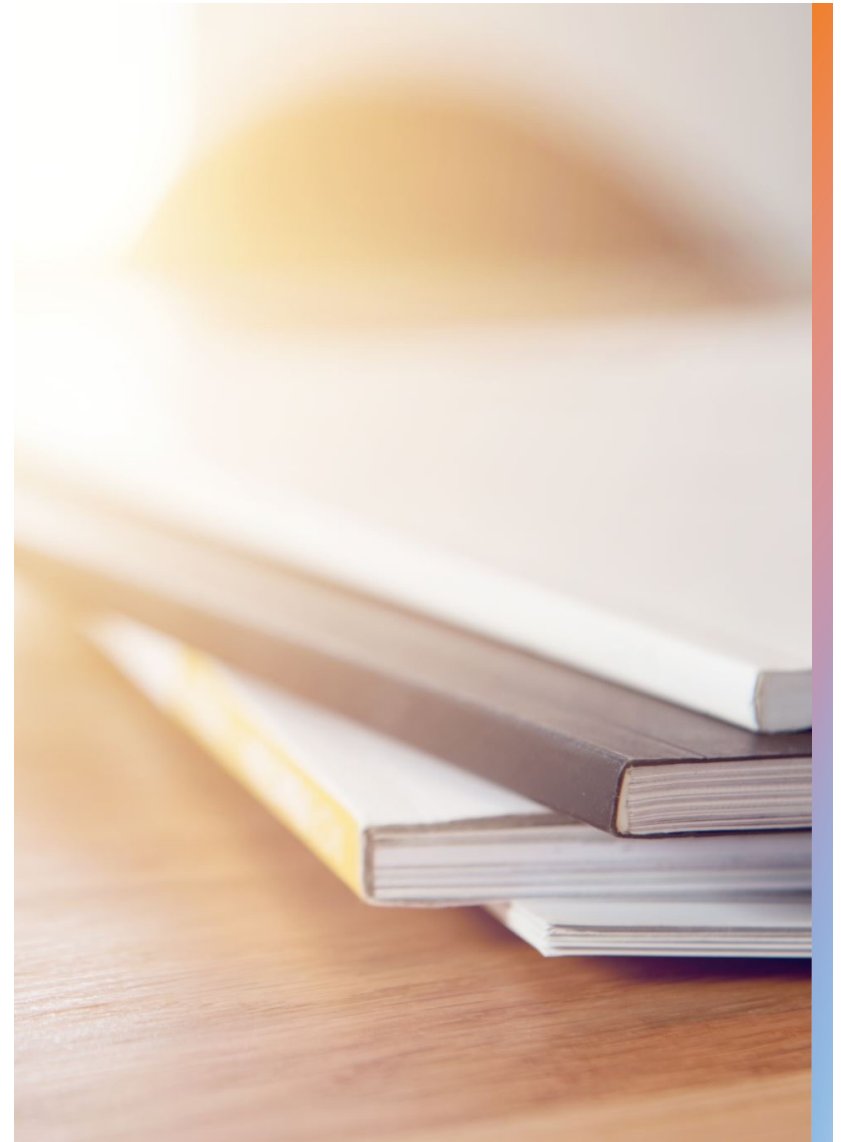
A lot of maths in the first few weeks.



If you choose this module, make sure that you are prepared (with knowledge, passion, persistency, etc)

Today's Content

- Module Information
- Contents of the module



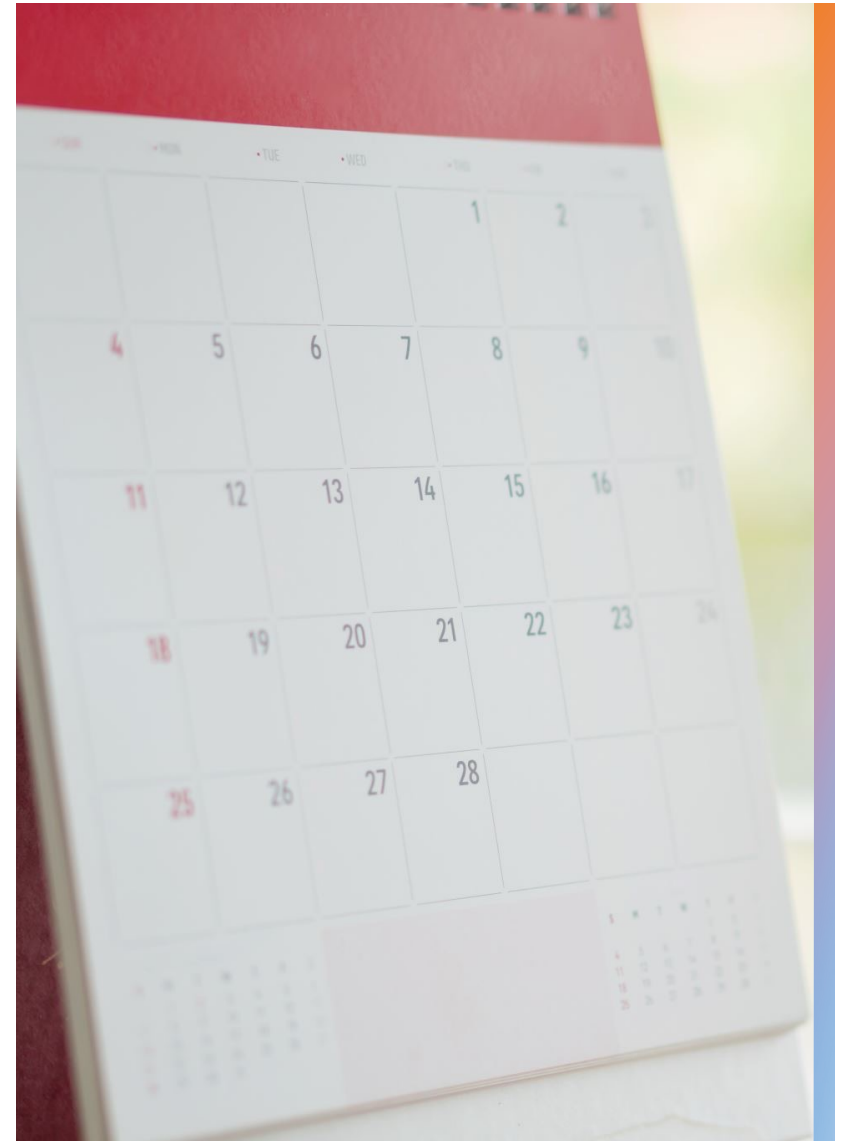
Module Outline

- The module consists of
 - 25~30 lectures
 - ~9 lab sessions
- Please ensure sufficient time on self study



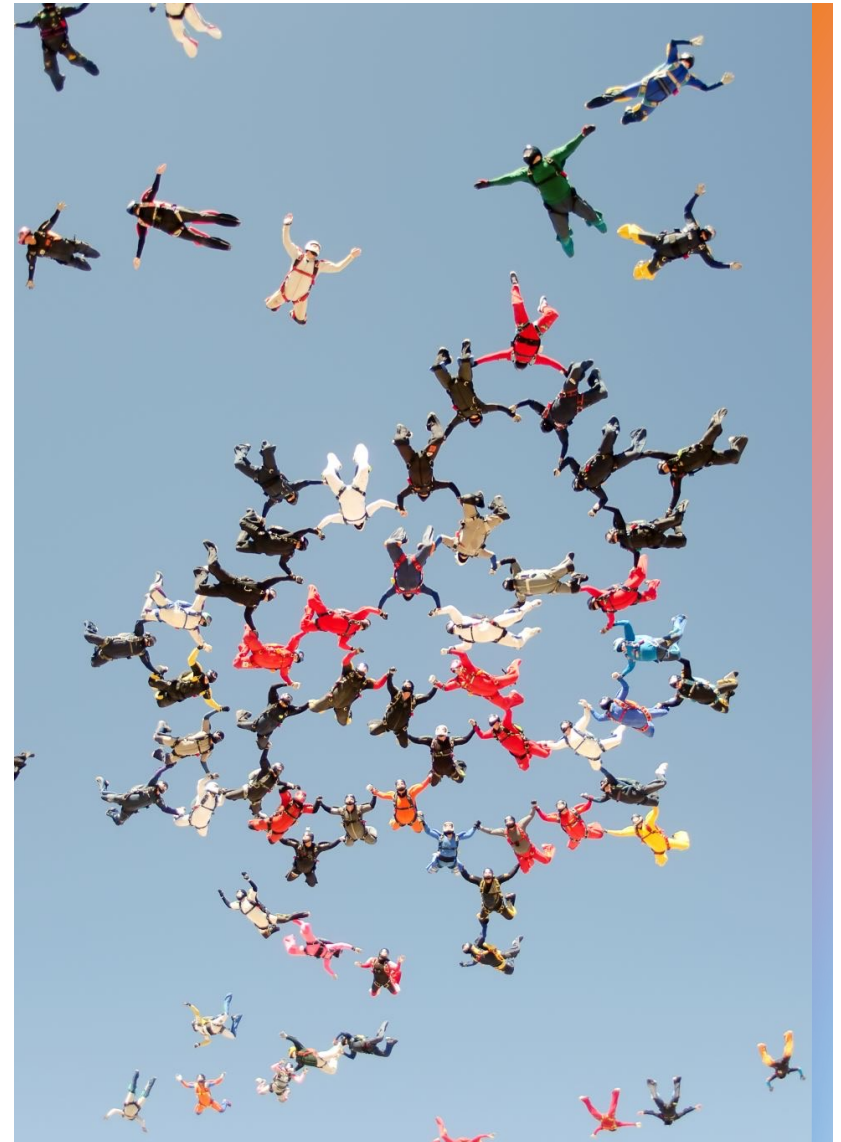
Module Outline

- Assessment
 - a 2-hour MCQ exam (70%)
 - two practical assignments (30%), submitted at the end of the semester. All submissions will attend an attack-defence competition.
- Module information on Canvas



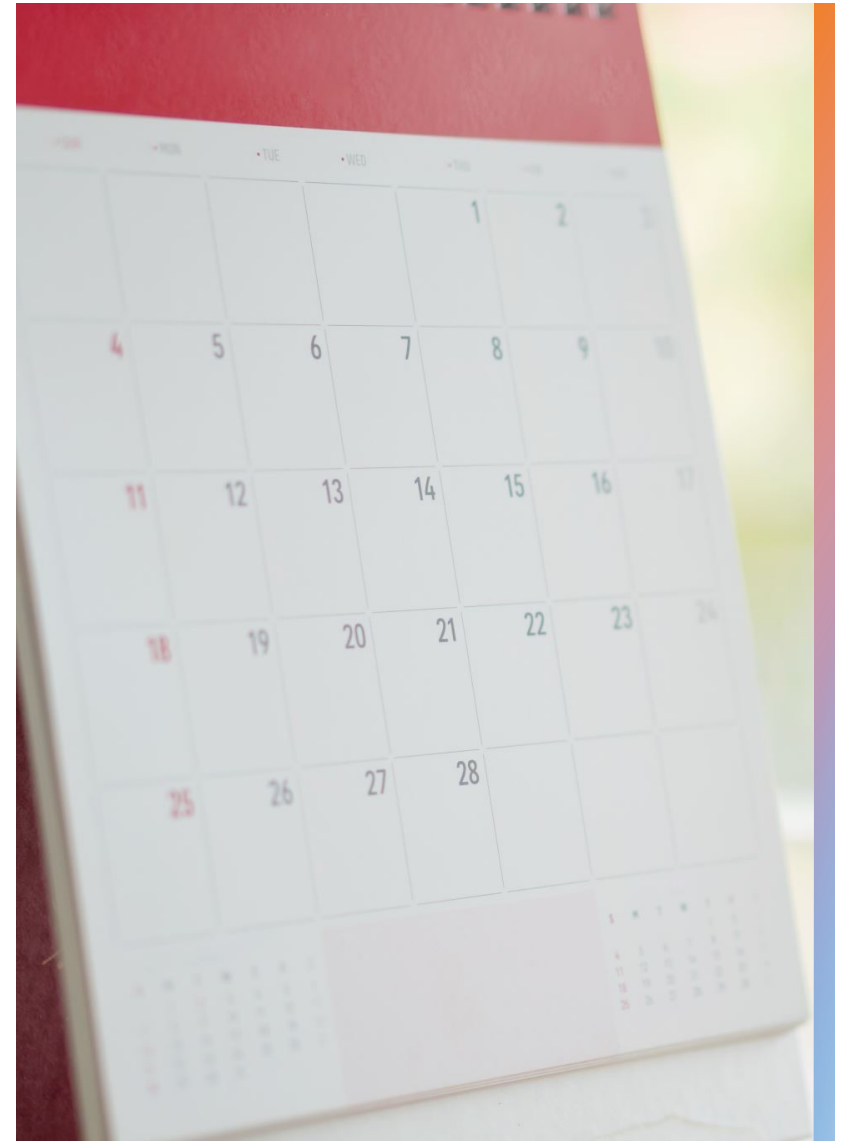
Module Delivery: Demonstrators

- 9 lab sessions
- About 200 students registered
- Who is going to support this?
 - Mr Gregory Stathopoulos
 - Mr Botond Hamori
 - Ms Lingfang Li



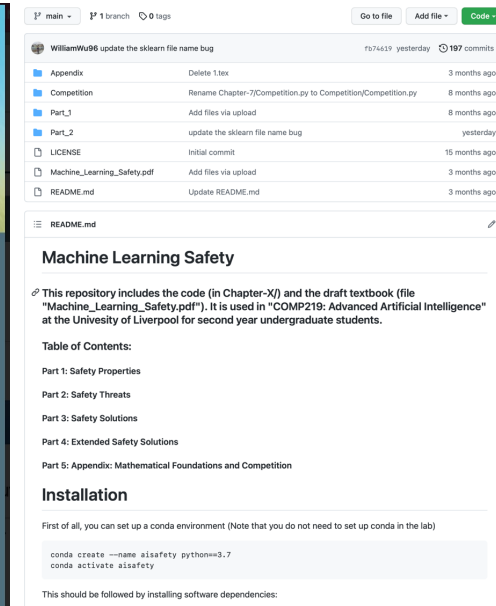
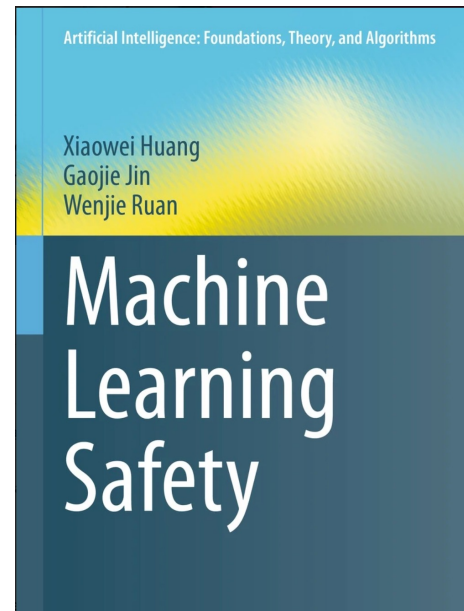
Timetable: Lectures

- Monday 16-17
- Tuesday 15-16
- Wednesday 12-13
- Slides will be distributed
- Will inform you through Canvas if I have to travel and cannot do lectures, so please watch your emails and canvas notification.

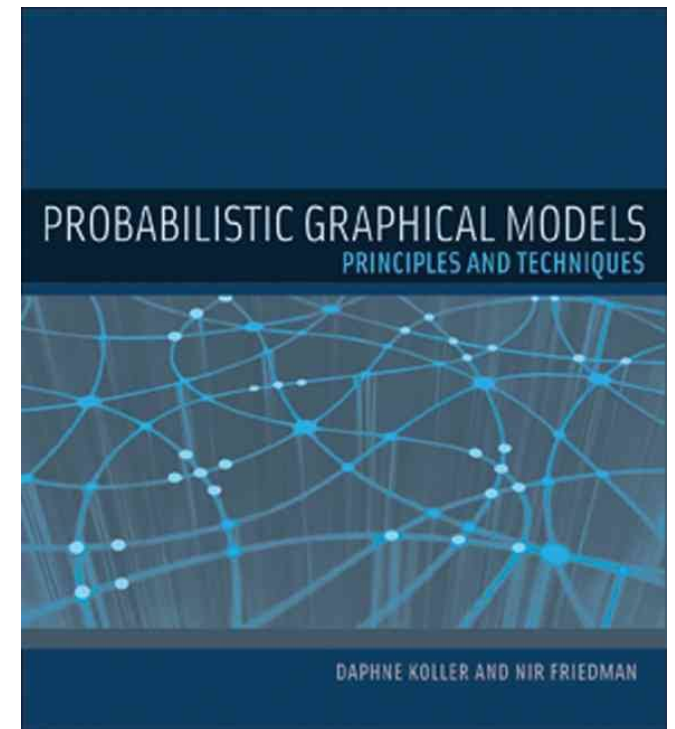
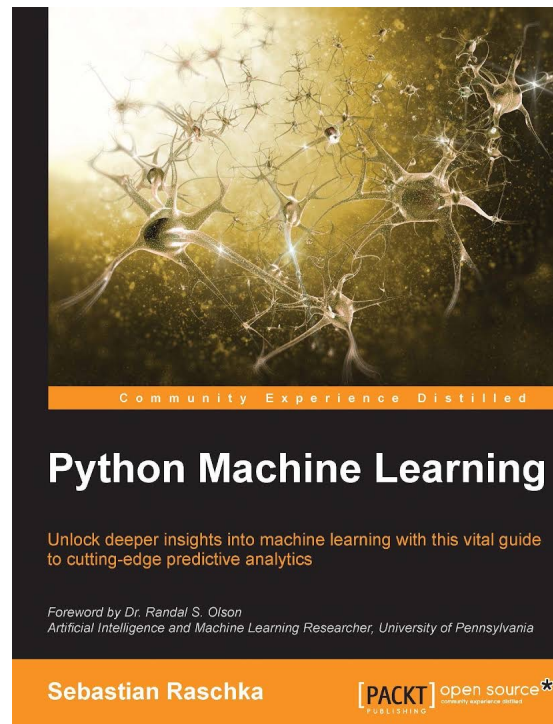
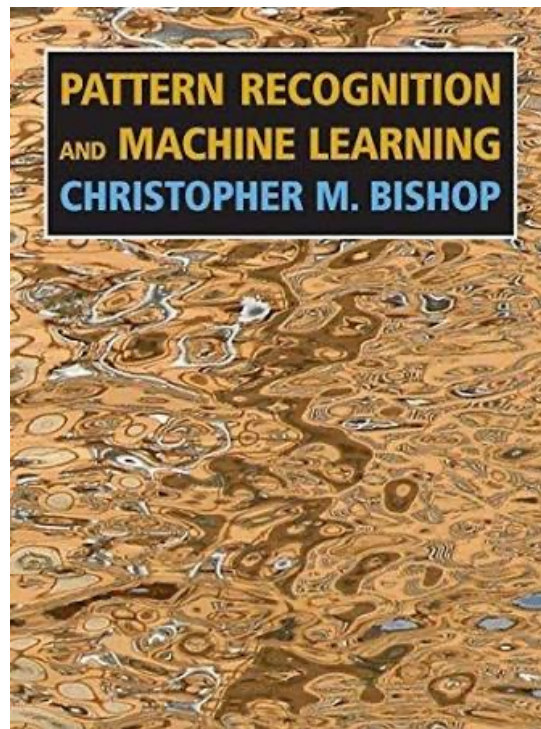


Teaching Materials

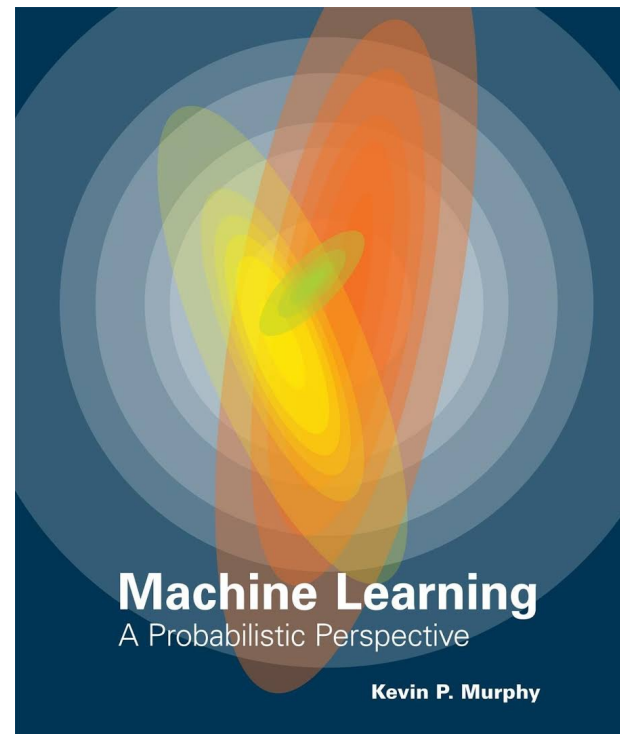
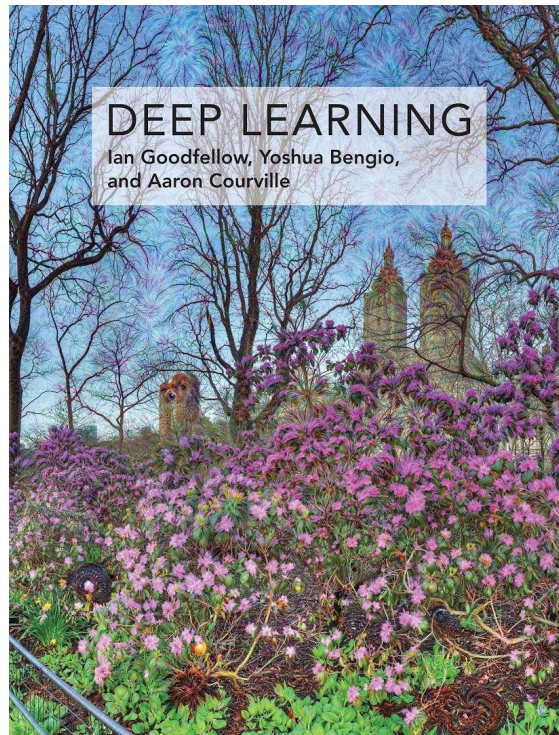
- <https://github.com/xiaoweiH/AISafetyLectureNotes>
 - Lecture notes, including all the theoretical knowledge you may need for this module
 - Lab practice, appears in “Practice” section of each chapter



Other Readings



Other Reading:



Python Coding

- **Essential** for assignments.
- Canvas has some basic materials as the starting point.



2. External Materials (NB: non-compulsory, for students interested in machine learning)

- Python and PyTorch Self-learning Materials. Below are a few pointers for those who want to get a better idea about python & pyTorch on deep learning:

- Python: <https://www.w3schools.com/python/>
- PyTorch: <https://pytorch.org/tutorials/>
- Python Tutorial Video: <https://www.youtube.com/watch?v=uQrJ0TkZlc&list=PLTjRvDozrdlxj5wgH4qkvwSOdHLOCx10f>



- PyTorch for Deep Learning: <https://www.youtube.com/watch?v=GlsG-ZUy0MY>



- Deep Learning with Python Videos: https://www.youtube.com/playlist?list=PLZbbT5o_s2xrfNyHZsM6ufI0iZENK9xgG

[How to learn deep learning by reading papers](#)

Week 3 -- [Decision tree and random forest](#)

Week 4 -- (for K-nn) [How to Search Data with KDTree](#)

Week 5 -- [Top 10 ML Algorithms Every Beginner Should Know](#)

Other Reading:



Tensorflow/PyTorch on-line documentations



Github (plenty of resources, code, tutorials, etc)



Various on-line courses



Reddit, quite some good discussions. Experts are around there.



Kaggle competitions, you can participate in to get more hand-on experience



Wikipedia, for various concepts, key pointers, etc



.... Many other on-line resources, please Google whatever you want

COMP111

- ▶ Brief history of AI including recent developments
- ▶ Intelligent Agents: A classification
- ▶ Search (applications: route planning, game playing)
- ▶ Knowledge Representation (applications: structured web search output)
- ▶ Reasoning under Uncertainty (application: almost everywhere)
- ▶ Learning (applications: face recognition, selfdriving cars)
- ▶ Philosophy and Ethics of AI (motivation: deducing sexual orientation from your picture ok? Visit <https://www.youtube.com/watch?v=Z3p3333333>)

Aims

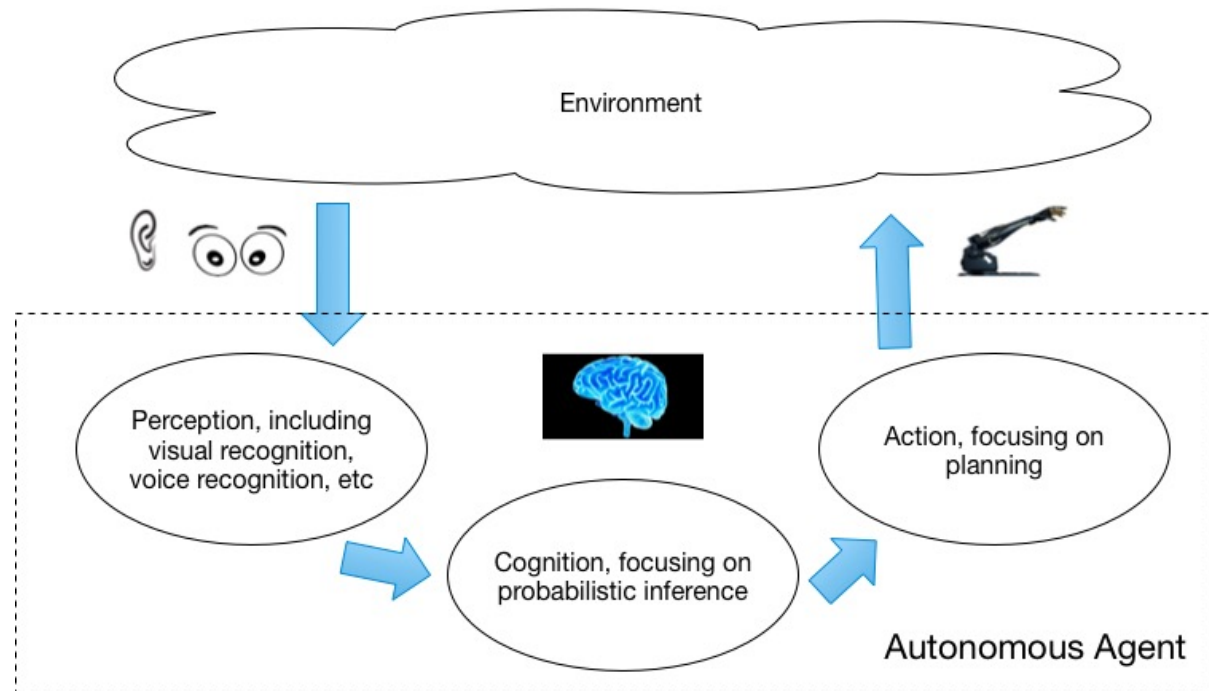
To equip students with the knowledge about basic algorithms that have been used to enable the AI agents to conduct the perception, inference, and planning tasks;

To equip students with the knowledge about machine learning algorithms;

To provide experience in applying basic AI algorithms to solve problems;

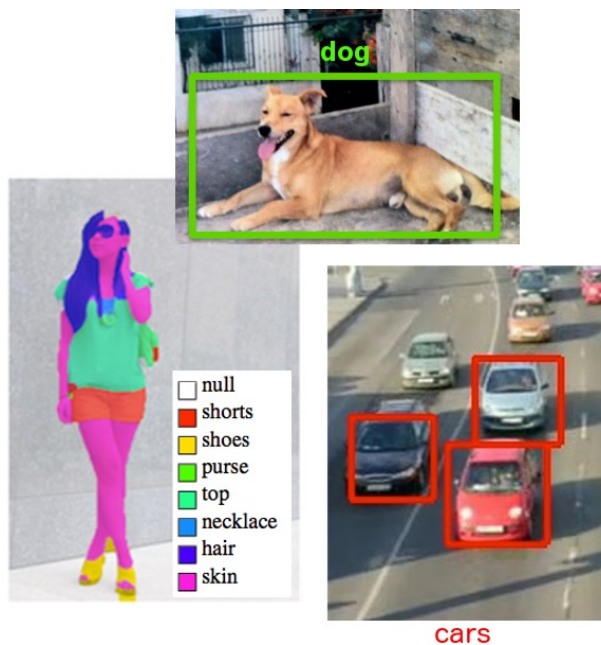
To provide experience in applying machine learning algorithms to practical problems;

Perception- Cognition- Action Loop



Perception

Teaching content:
traditional learning,
deep learning



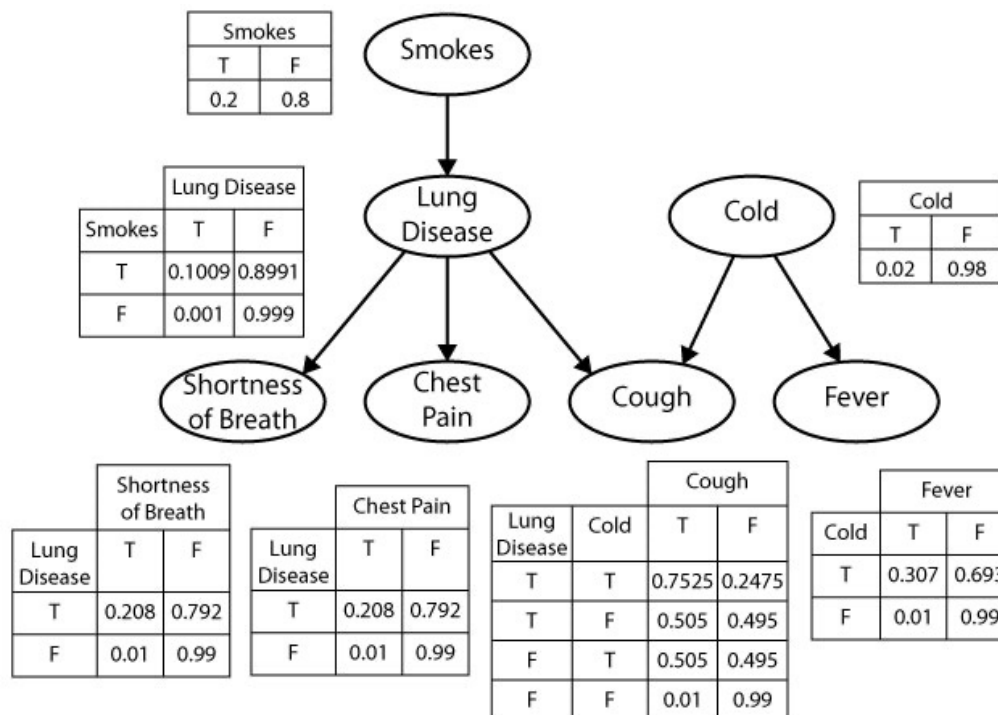
Visual Recognition



Voice Recognition

...

Cognition by Probabilistic Inference

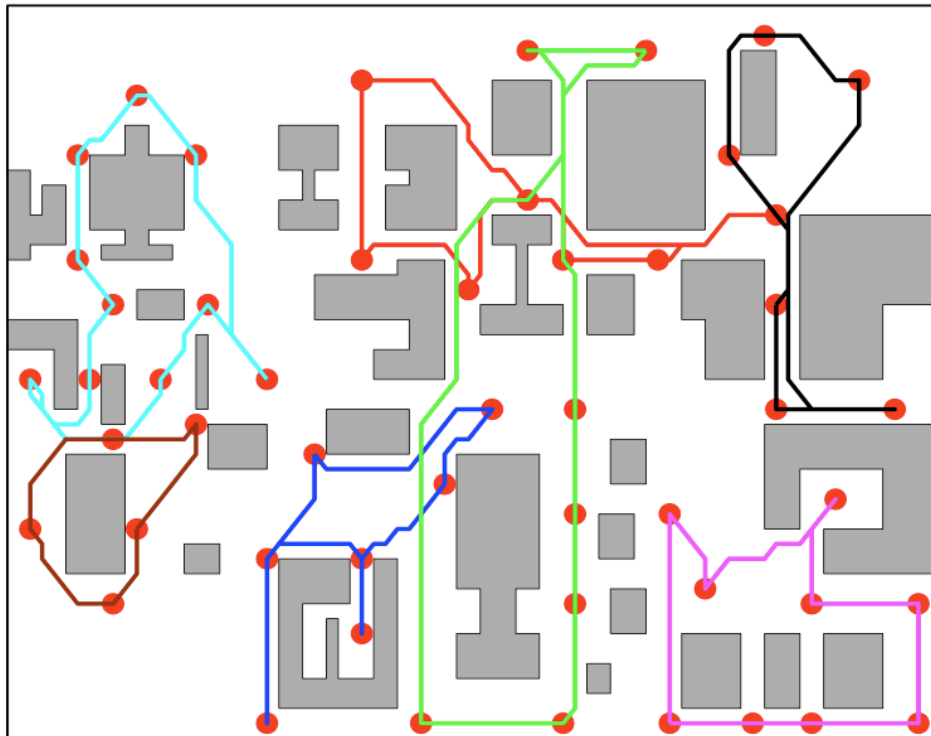


Q. how to automatically
infer the disease (e.g.,
lung disease, cold, etc)
from the symptoms (e.g.,
smokes, shortness of
breath, chest pain,
cough, fever, etc)?

Note: Symptoms obtained
from perception.

Teaching content:
in other modules, e.g., COMP111,
COMP222

Action by Planning



After cognition, we may use
the obtained knowledge to
react to the environment

Q: in the factory floor as
shown in the left diagram,
how many robots is needed
to patrol the area? and **how**
to plan their activities?

Learning Outcomes



Ability to explain in detail how the techniques in the perceive-inference-action loop work



Ability to choose, compare, and apply suitable basic learning algorithms to simple applications



Ability to explain how deep neural networks are constructed and trained, and apply deep neural networks to work with large scale datasets



Ability to conduct probabilistic inference.